SSH authentication analysis

Threat hunting and brute-force detection using Splunk Enterprise

TOTAL EVENTS	FAILED	SUCCESSFUL
2,400	1,216	612

Executive summary

This investigation reviews JSON-based Zeek-style SSH events in a local Splunk Enterprise instance. The analysis examines authentication outcomes, repeated failures, source hosts, destination servers, and reusable detection logic.

PRIORITY SOURCE

10.0.0.25

78 events | 174 failed authentication attempts | maximum 8 per event

The source produced 18 failed-login events and 26 multiple-failure events, alongside 16 successful logins and 18 connections without authentication. This pattern supports investigation of potential brute-force behavior; successful events alone do not establish compromise.

Investigation	Value
Analyst / date	Majed Almabadi / June 19, 2026
SIEM / deployment	Splunk Enterprise / local instance
Data / index	ssh_logs.json / auth_logs
Authentication not completed	572 events (auth_success = null)

Data and investigation plan

Environment, extracted fields, and investigation sequence

Environment	Configuration
Operating system	Ubuntu 26.04 LTS
Access	Splunk Web Interface
Source	30-Days SOC Challenge Beginner - Day #18
Format	JSON-formatted Zeek-style SSH logs

Dataset fields

Field	Meaning
ts / uid	Event timestamp / unique session identifier
id.orig_h / id.orig_p	Source host IP address / source port
id.resp_h / id.resp_p	Destination IP address / port (usually 22)
proto	Transport protocol
auth_success	Authentication outcome: true, false, or null
auth_attempts	Number of authentication attempts
event_type	Human-readable event classification

Investigation sequence

1. **Ingest:** upload JSON and verify searchability.
2. **Discover:** validate source, destination, outcome, attempts, and event fields.
3. **Baseline:** compare event categories and authentication outcomes.
4. **Investigate:** identify sources with repeated failed attempts.
5. **Reconstruct:** review suspicious activity chronologically.
6. **Detect:** prepare an SPL search for a scheduled alert.

Baseline authentication activity

Searches 01-03 / ingestion, event categories, and outcomes

01 / Verify ingestion

```
index=auth_logs
| stats count
```

The auth_logs index returned **2,400 events**, confirming that the SSH dataset was searchable.

02 / Discover event categories

```
index=auth_logs
| stats count by event_type
| sort -count
```

The event_type field separates successful logins, failed logins, multiple failed authentication attempts, and connections without authentication.

Event category	Count
Successful SSH Login	612
Failed SSH Login	610
Multiple Failed Authentication Attempts	606
Connection Without Authentication	572

03 / Compare outcomes

```
index=auth_logs
| stats count by auth_success
```

The report records **1,216 false**, **612 true**, and **572 null** outcomes. Failed-login and multiple-failure categories together account for the 1,216 failed events. Null is distinct from false; review event_type alongside this search to retain unauthenticated activity.

Source and destination hosts

Searches 04-05 / highest-volume SSH endpoints

04 / Identify source hosts

```
index=auth_logs
| stats count by id.orig_h
| sort -count
```

Source **10.0.0.25** was the most active source in the recorded results, with **78 events**.

id.orig_h	count
10.0.0.25	78
10.0.0.18	64
10.0.0.21	64
10.0.0.48	64
10.0.0.14	62
10.0.0.44	62
10.0.0.57	62
10.0.0.16	60

05 / Identify destinations

```
index=auth_logs
| stats count by id.resp_h
| sort -count
```

Destination hosts **10.0.1.2** and **10.0.1.6** each received **230 events**. The incident summary separately identifies **10.0.1.10** as the target for the highlighted incident.

id.resp_h	count
10.0.1.2	230
10.0.1.6	230
10.0.1.9	226
10.0.1.4	218
10.0.1.10	208
10.0.1.1	198
10.0.1.12	198
10.0.1.7	186

Candidates and source timeline

Searches 06-07 / repeated failures and suspicious-source activity

06 / Filter brute-force candidates

```
index=auth_logs auth_success=false
| where auth_attempts >= 5
| table _time uid id.orig_h id.resp_h auth_attempts event_type
| sort -auth_attempts
```

Failed authentication events with **auth_attempts >= 5** were treated as brute-force candidates. This is a per-event attempt threshold, rather than five separate events in a defined interval.

07 / Reconstruct the timeline

```
index=auth_logs id.orig_h="10.0.0.25"
| sort _time
| table _time uid id.orig_h id.orig_p id.resp_h id.resp_p
        auth_success auth_attempts event_type
```

The timeline returned **78 events** from 10.0.0.25. Failed authentication activity totaled **174 attempts**, with a highest single-event value of **8**.

Source event category	Events
Multiple failed authentication attempts	26
Failed SSH login	18
Connection without authentication	18
Successful SSH login	16
Total source events	78

Investigate successful logins in the context of the failed activity. These source-level totals describe the recorded investigation; the dataset is not included in this repository.

Detection and incident findings

Search 08 / reusable alert logic and analyst assessment

08 / Prepare alert logic

```
index=auth_logs auth_success=false
| where auth_attempts >= 5
| stats max(auth_attempts) as max_attempts count
  by id.orig_h id.resp_h
| where max_attempts >= 5
| sort -max_attempts
```

Save this search as a scheduled alert for repeated failed SSH authentication candidates. The source report does not specify a schedule, lookback, or suppression window; configure these for the deployment.

Incident detail	Recorded value
Suspicious source / target	10.0.0.25 / 10.0.1.10
Peak attempts / result	8 / false
Event / incident time	Multiple Failed Authentication Attempts 2025-04-24 10:20:09 UTC

Findings and severity

Finding	Evidence	Severity
Suspicious source activity	78 source events	Medium
Repeated failures	18 failed-login + 26 multiple-failure events	High
Possible brute force	174 failed attempts; maximum 8 per event	High
Successful logins	16 from the same source	Medium
No completed authentication	18 source connection events	Low

The evidence is consistent with possible brute-force or repeated unauthorized SSH access attempts. Monitor the source and investigate its successful logins.

Response and recommendations

Controls, conclusion, and investigation references

Control	Purpose	Priority
Disable direct root SSH login	Reduce guessing against privileged accounts.	High
Strong passwords or SSH keys	Reduce successful password attacks.	High
MFA where supported	Add a factor beyond passwords.	High
Fail2Ban or lockout rules	Block repeated failed attempts.	Medium
Monitor SSH logs in Splunk	Improve detection and response visibility.	High
Scheduled Splunk alert	Automate threshold-based detection.	High

Conclusion

The investigation identified 10.0.0.25 as a suspicious SSH source based on repeated failed authentication activity, successful logins from the same source, and a maximum auth_attempts value of 8. Reusable SPL searches support repeatable investigation and future alerting.

References

Splunk Search Reference

Splunk - Get Data In

MITRE ATT&CK; T1110 - Brute Force

OWASP - Brute Force Attack

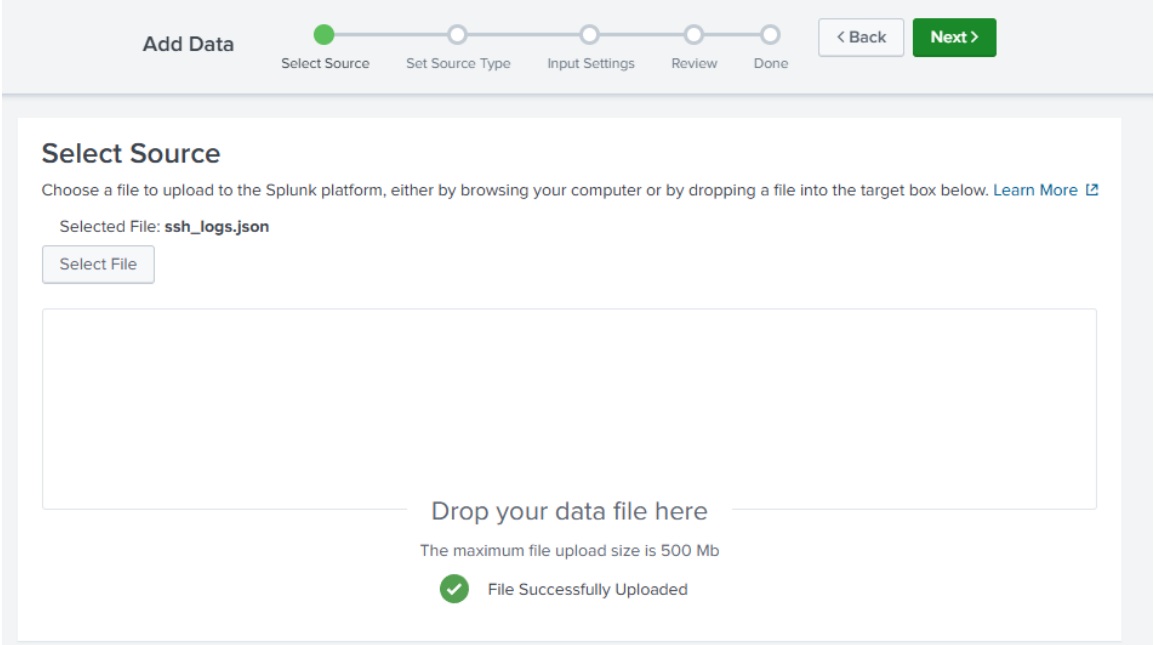
Dataset source - SOC Challenge Day #18

Original Splunk screenshots are retained on the following pages. README dashboards visually reconstruct the documented results.

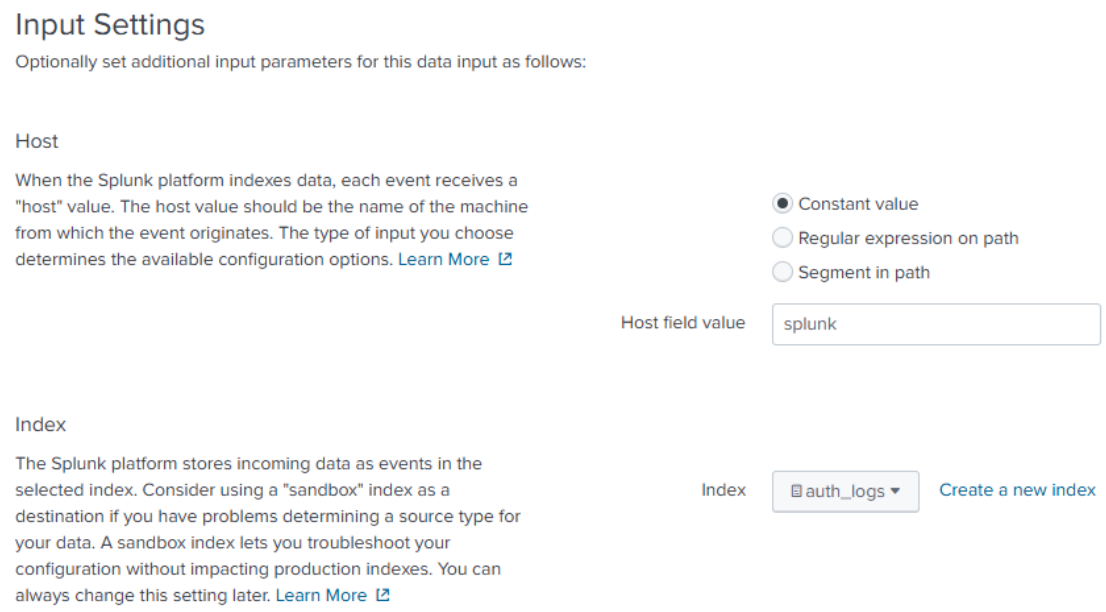
Lab evidence / data ingestion

Original Splunk screenshots from the investigation

Upload the SSH JSON file



Configure host and destination index



Lab evidence / search results

Original Splunk tables from the investigation

Ingestion count

count
2400

Authentication outcomes

auth_success	count
false	1216
null	572
true	612

Event category counts

event_type	count
Successful SSH Login	612
Failed SSH Login	610
Multiple Failed Authentication Attempts	606
Connection Without Authentication	572

Brute-force candidate search

_time	uid	id.orig_h	id.resp_h	auth_attempts	event_type
2025-04-24 10:20:09.518	SH4068174	10.0.0.19	10.0.1.1	8	Multiple Failed Authentication Attempts
2025-04-24 10:20:09.518	SH5439493	10.0.0.15	10.0.1.7	8	Multiple Failed Authentication Attempts
2025-04-24 10:20:09.518	SH9535038	10.0.0.24	10.0.1.10	8	Multiple Failed Authentication Attempts
2025-04-24 10:20:09.518	SH7403822	10.0.0.18	10.0.1.3	8	Multiple Failed Authentication Attempts
2025-04-24 10:20:09.518	SH4845525	10.0.0.56	10.0.1.9	8	Multiple Failed Authentication Attempts
2025-04-24 10:20:09.518	SH5536175	10.0.0.13	10.0.1.9	8	Multiple Failed Authentication Attempts
2025-04-24 10:20:09.518	SH5348995	10.0.0.57	10.0.1.6	8	Multiple Failed Authentication Attempts